

(Graduate) **Approximation Algorithms CSE 269**

Instructor: Vaggos Chatziafratis (Fall 2025)

COURSE INFORMATION

This is a **graduate** course on **Approximation Algorithms**, meant for graduate students with a good mathematical background. Students would need to be familiar with discrete math, analysis of algorithms, basic analysis, probability theory, and graph theory.

The class is aimed at students who want to learn more about the central ideas on how to deal with NP-Hard problems that are ubiquitous in today's data science applications.

The class will focus on proving performance guarantees for many well-known algorithms as well as their practical applications and implementation details (how and when to apply them). Topics include combinatorial algorithms, clustering, linear programming, LP duality, semidefinite programming and their duals, algorithms for SAT, tree metrics, randomized rounding, dimensionality reduction, sum of squares, metric embeddings, and gradient descent.

The course will cover the key tools of probabilistic analysis and optimization necessary, but this is a theory-heavy class, so many concepts from mathematics, calculus, probability will be assumed.

In Short:

This proof-based class is about Approximation Algorithms (meant for graduate students) and how these tools are used in state-of-the-art algorithms for many important NP-hard problems. Topics: clustering, linear programming, LP duality, semidefinite programming and their duals, algorithms for SAT, tree metrics, randomized rounding, dimensionality reduction, metric embeddings, and gradient descent.

INSTRUCTOR INFORMATION

Vaggos Chatziafratis (vaggos@ucsc.edu)

LEARNING OUTCOMES

This course is a graduate introduction to approximation algorithms, a hot topic in algorithms, used in a variety of real-world applications.

Students will learn a lot of state of the art research development and they will be able to read papers, and be able to read advanced textbooks in theoretical computer science, and do more novel research in the field.

- Students will get the basic tools of the trade to do research in theoretical computer science and theoretical machine learning (e.g., clustering is a cornerstone of ML as

it is unsupervised learning/pattern recognition). This will aid their ability to apply reasoning, problem solving, and technical skills to solve problems with minimal guidance, and to conduct independent and innovative research (one of the CSE PhD PLOs).

- Students will get a significantly better understanding of how theoretical concepts can be applied in real-world settings and have important applications and impact. For example they will learn how vast amounts of data can be efficiently searched, can be efficiently stored or compressed without loss in information (e.g., through the fundamental Johnson-Lindenstrauss embedding theorem).
- Students will familiarize themselves with concepts used throughout the curriculum such as probability, linear algebra, algorithmic techniques for analyzing datasets, all of which could come in handy later in their studies.
- Students are expected to have basic knowledge around algorithms (having taken before an undergraduate class in Algorithms) and this is a theory-focused course.

PREREQUISITES

Enrollment is restricted to graduate students or by permission of the instructor. Students taking this course need to have had at least one prior course in algorithms, similar to CSE 102 or equivalent. Students will need a solid background in analysis of algorithms, discrete math, probability theory, graph theory, and overall mathematical maturity.

REQUIRED MATERIALS, TEXTBOOKS AND TECHNOLOGY

- Shmoy's and Williamson's Approximation Algorithms book:
<https://www.designofapproxalgs.com/book.pdf>
- Vazirani's Approximation algorithms book:
<https://ics.uci.edu/~vazirani/book.pdf>
- Randomized Algorithms by Motwani and Raghavan.
<https://www.cambridge.org/core/books/randomized-algorithms/6A3E5CD760B0DDBA3794A100EE2843E8>
- Probability and Computing by Mitzenmacher and Upfal
<https://www.cambridge.org/core/books/probability-and-computing/3A5B47DB315FC64B9256C5C8131C5EFA>

COMMUNICATION

ASSIGNMENTS & ASSESSMENT

- **Research Project and Presentation:** I will post several papers to read and the students will need to come up with a 1-2 papers they must read, and a more research level question, and study a novel problem in theoretical computer science related to what they read. Students will do a mini-presentation in class for the paper they read (and ideally, how they extended it). Depending on how many students are in class I will allocate time towards Week 6th onwards. Novel results will receive top scores.
- **Deliverables for Research Project:** 10-page pdf report with background, and results. If people want to work in groups of 2 (max 3 people) we must coordinate early in the class by the end of Week 3.
- **Grade:** Grade is calculated based on 70% Research Project and Presentation, 30% Final.
- **Spot-Checks:** At any point, you should be in position to explain your solutions and your writings. Any writing should be your own.

STUDENT FEEDBACK

At the end of the quarter you will be asked to complete a Student Experience of Teaching survey for this course. SETs provide an opportunity for you to give valuable feedback on your learning that is honest and constructive. This anonymous feedback will help me consider modifications to the course that will help future students learn more effectively.

COURSE SCHEDULE

Week	Reading and Activities	Notes
One	Approximation and how to deal with NP-Hardness, Started discussing hierarchical clustering objectives	None this week
Two	Use of SDPs for clustering and graph partitioning problems	List of papers Send me top 3 by October 13th
Three	Max Cut, Correlation Clustering	Max Cut Paper , Correlation Clustering

Four	SDP for Betweenness	Paper , Linear-time algo ,
Five	Beyond Random Performance for Hierarchical clustering and Betweenness	Paper
Six	Ordinal Embeddings proofs	Terminal Embeddings Ordinal Embeddings Indyk's paper for diameter
Seven	Embedding Dimension for Ordinal Constraints	Paper
Eight	Presentations	
Nine	Presentations	
Ten	Presentations ID: 752 731 7535 (passcode 555999) https://ucsc.zoom.us/j/7527317535?pwd=QTVjekNKSGU3R1R2YWVibEJrciNqdz09	Paper for Monday lecture for local optimization hardness, Notes from Yiyuan's lecture
Final Week + Final Report	Monday, Dec 8 (take home exam, submit by email your responses, to be posted here) Friday, Dec 12 (submit final pdf report)	

ACADEMIC INTEGRITY per UCSC regulations

All members of the UCSC community benefit from an environment of trust, honesty, fairness, respect, and responsibility. You are expected to present your own work and acknowledge the work of others in order to preserve the integrity of scholarship.

Academic integrity includes:

- Following exam rules
- Using only permitted materials during an exam
- Viewing exam materials only when permitted by your instructor
- Keeping what you know about an exam to yourself
- Incorporating proper citation of all sources of information
- Submitting your own original work

Academic misconduct includes, but is not limited to, the following:

- Disclosing exam content during or after you have taken an exam
- Accessing exam materials without permission
- Copying/purchasing any material from another student, or from another source, that is submitted for grading as your own
- Plagiarism, including use of Internet material without proper citation
- Using cell phones or other electronics to obtain outside information during an exam without explicit permission from the instructor
- Submitting your own work in one class that was completed for another class (self-plagiarism) without prior permission from the instructor.
- Violations of the Academic Integrity policy can result in dismissal from the university and a permanent notation on a student's transcript. For the full policy and disciplinary procedures on academic dishonesty, students and instructors should refer to the [Academic Misconduct page](#) at the [Division of Undergraduate Education](#).